

Milestone 2 — Complete Support Workflow and Product Readiness

1. Business Context

The application foundation is ready.

The company now wants customers and support employees to manage requests through the platform instead of relying on external communication channels.

The full business process must work across both applications using the same data. It must also remain reliable during normal daily use, including slow networks, failed updates, large request lists, file attachments, changing data, and smaller screens.

2. Business Goals

- Allow customers to create and follow support requests.
- Allow agents to find and handle support work.
- Keep ownership and urgency clear.
- Keep customer communication inside the request.
- Keep internal support communication private.
- Allow managers to review and reassign work.
- Keep the experience usable as request volume grows.
- Keep the interface accurate after failed actions.
- Keep users informed when data changes.
- Support attachments safely.
- Give managers practical operational visibility.
- Improve responsiveness, accessibility, and test coverage.
- Make the repository easy for another developer to run.

3. Customer Experience — React

A customer should be able to:

- submit a support request,

- choose a category and urgency,
- view only their own requests,
- understand whether a request is active, waiting, or completed,
- open a request,
- review the conversation,
- send additional information,
- review the final resolution.

4. Support Experience — Angular

A support employee should be able to:

- view requests requiring attention,
- identify urgent requests,
- identify unassigned requests,
- open a request,
- take responsibility for an available request,
- communicate with the customer,
- add internal support notes,
- update progress,
- complete the request.

A support manager should additionally be able to:

- review the full workload,
- reassign responsibility.

5. Shared Business Process

1. The customer submits a request.
2. The request becomes visible to the support team.
3. A support employee becomes responsible for it.
4. The issue is investigated.
5. Customer and support communication takes place inside the request.
6. Internal notes remain available only to support employees.
7. The request is resolved and completed.
8. A completed request may become active again when further support is required.

6. Core Workflow Requirements

- Every request must have a clear reference, description, category, urgency, status, customer, owner when assigned, and activity information.
- Customers must only access their own requests.
- Internal notes must never be visible to customers.
- Support employees must be able to find work that requires attention.
- Available requests may be claimed by an agent.
- Managers may reassign responsibility.
- Invalid actions for the current request state must be prevented or handled safely.
- Completed requests should not accept normal communication unless they become active again.
- Important actions should communicate loading, success, failure, validation, and empty states clearly.
- The support workspace should support practical search, filtering, sorting, and pagination.
- Normal browsing should not require loading every request into the browser.

7. Product Readiness Requirements

Attachments

Users may need to attach files to a support request.

The solution should define:

- who may upload,
- who may view,
- which files are accepted,
- how invalid files are handled,
- how protected attachments remain protected.

Update Feedback

The interface should feel responsive when users make changes.

If a change fails, the interface must not leave the user believing the operation succeeded.

Current Data

Users should receive reasonably current request information.

The implementation may use realtime updates or another appropriate approach.

New updates must not create duplicate or contradictory information in the interface.

Manager Visibility

Managers should have a practical summary of support activity.

Basic summary information should not require loading the entire request dataset into the browser.

Responsive Experience

The main customer and support workflows must remain usable on smaller screens.

Accessibility

Important interactive workflows should support reasonable keyboard use and semantic interaction.

8. Business Constraints

- Protected information must remain protected even if a user attempts direct access.
- Data shown after an update should remain consistent with the shared backend state.
- The customer experience should use customer-friendly language.
- Both applications must represent the same request lifecycle.
- Existing customer data isolation must remain enforced.
- Internal notes must remain private.
- Failed updates must not leave misleading UI state.
- Performance improvements must not change business behavior.
- File access must respect the same business permissions as request access.

9. Delivery Expectations

Before implementation, provide:

- clarification questions,
- proposed request lifecycle,
- assumptions around ownership,
- assumptions around when customers may reply,
- proposed approach for searching, filtering, and browsing growing request data.

The final milestone delivery should include:

- the complete shared workflow,
- local setup instructions,
- backend/data-service setup,
- safe environment configuration examples,
- documented test users,
- appropriate automated tests,
- assumptions and known limitations,
- workflow demonstration instructions.

10. Review Preparation

Be ready to demonstrate:

1. Customer creates a request.
2. Agent finds and claims it.
3. Agent sends a customer-visible message.
4. Agent adds an internal note.
5. Customer replies.
6. Agent completes the request.
7. Customer attempts to access data they do not own.
8. Manager reassigns the request.

Be ready to explain:

- where server data is managed,
- what happens when an update fails,

- how customer data remains isolated,
- how internal notes and attachments remain protected,
- how growing request lists are handled,
- how current/live updates are reconciled,
- which parts of the interface are most performance-sensitive,
- which automated tests provide the most confidence.

11. Optional Business Enhancements

Only after the required product is complete:

- Saved filters.
- Live notifications.
- Customer satisfaction rating.
- Advanced keyboard navigation.
- Additional manager analytics.